Dynamic Transport and Flows for ML

Eulerian PDEs, Lagrangian particles, Wasserstein gradient flows, and transportation views of modern ML.

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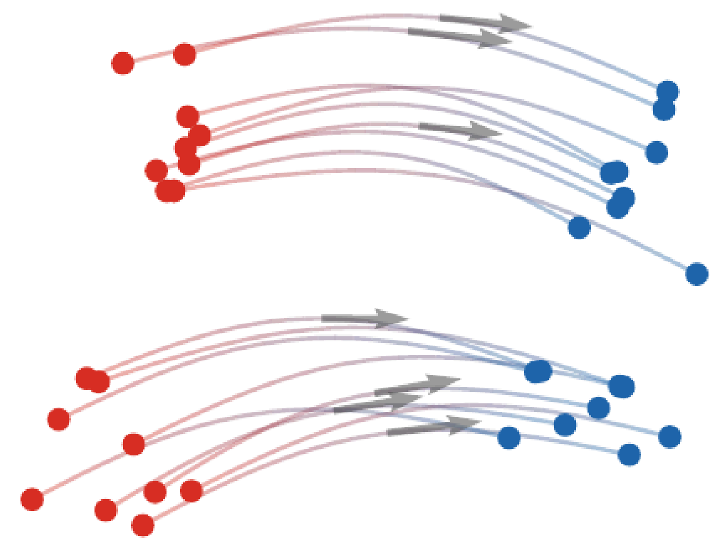
Optimal Transport for Machine Learners

Why dynamic flows?

Machine learning increasingly manipulates distributions by evolving them:

- diffusion and flow-matching generative models,
- gradient-flow training dynamics,
- population dynamics in neural networks,
- token dynamics in transformers,
- biological trajectories and multi-omics integration.

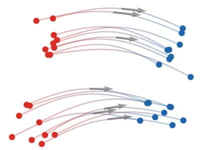
Dynamic OT gives a common language for these evolutions.



A model can generate by transporting a simple law toward data.

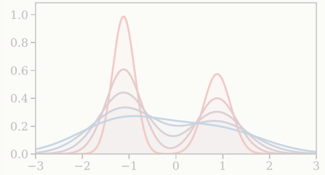
Roadmap: generative transport

1. Generative transport



2. Dynamic OT

3. Gradient flows



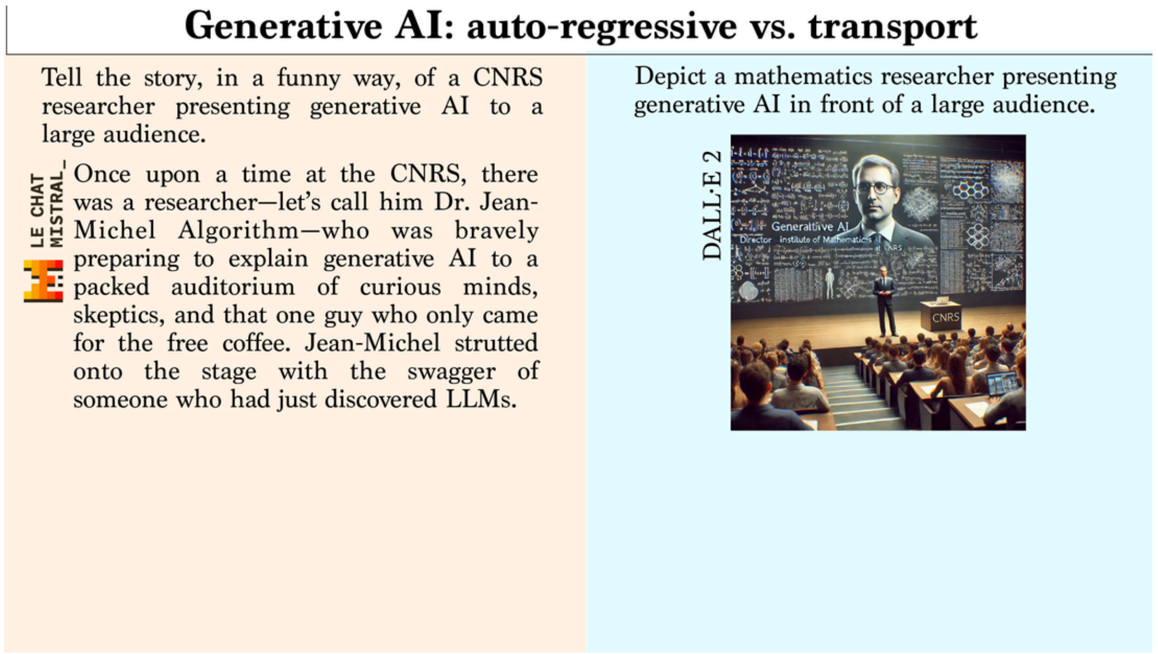
4. Training and biology

5. Attention dynamics

Auto-regressive models versus transport

Auto-regressive models generate one token at a time. Transport models instead evolve a whole distribution or point cloud continuously.

Transport viewpoint.
Generation is the construction of a map or flow sending a simple reference law α_0 to a data law α_1.



Eulerian and Lagrangian descriptions

Two equivalent views.
A curve of measures $(\alpha_t)_{t\in[0,1]}$ can be described:

- in Eulerian form by a density and velocity field (ρ_t, v_t),
- in Lagrangian form by particle trajectories $\dot{X}_t = v_t(X_t)$.

When $\alpha_t = \rho_t\,dx$, the velocity v_t is an Eulerian field on space; when X_t is random, $\alpha_t = \mathrm{Law}(X_t)$.

The two descriptions are linked by the continuity equation

$$\partial_t \rho_t + \mathrm{div}(\rho_t v_t) = 0.$$

Gradient structure links the two views

This is the bridge from kinematics to optimization: the metric selects the velocity field.

Wasserstein gradient.
For a functional $\mathcal{F}$, the steepest-descent velocity is

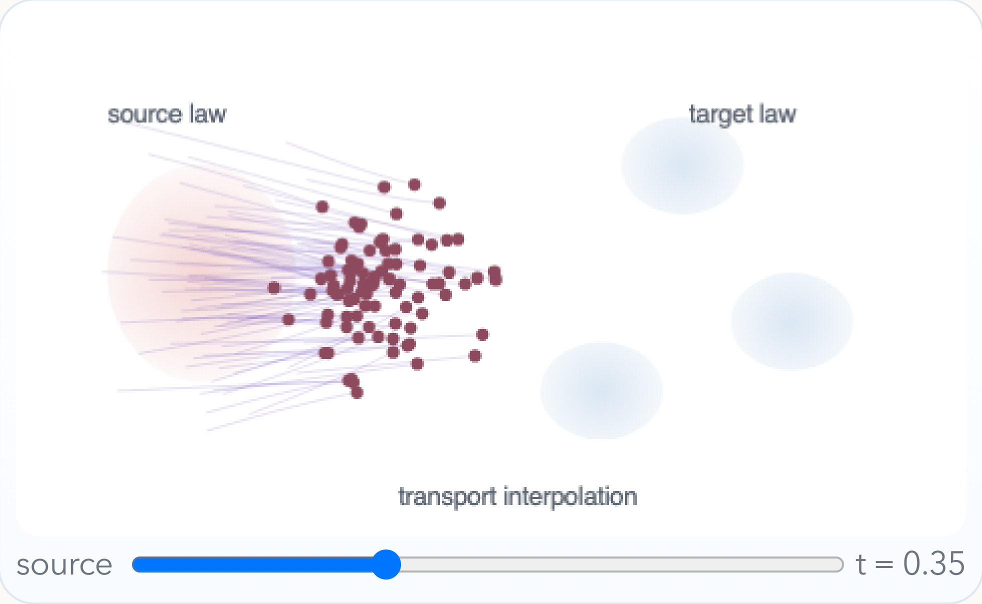
$$v_t = -\nabla \frac{\delta \mathcal{F}}{\delta \rho}(\rho_t), \qquad \partial_t \rho_t + \mathrm{div}(\rho_t v_t) = 0.$$

The Eulerian PDE and the Lagrangian particle ODE are two descriptions of the same gradient flow.

This viewpoint is reused for sampling, neural-network training, genomics, and attention dynamics.

Interactive: transport sampling

A deterministic transport sampler moves particles from a simple source cloud to a target cloud. The slider shows a Lagrangian interpolation.



Live panel: transport flow sampling

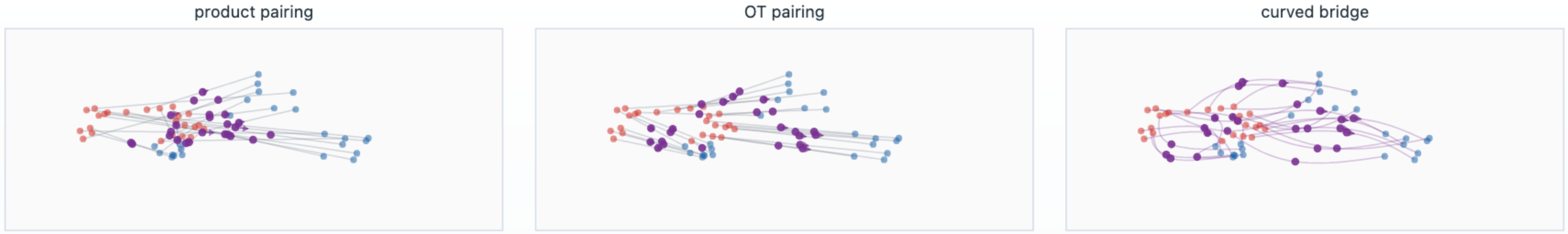
particles
24
24

time
0.50
0,50

curvature
0.32
0,32

seed
3606
3606

Reset
Open larger



24 paired particles; time 0.50; curved-bridge bend 0.32

The MyST panel exposes the same sampler with the book controls and notation.

Stochastic interpolants and flow matching

Interpolant.
A stochastic interpolant defines random variables

$$X_t = I_t(X_0, X_1, Z), \qquad X_0 \sim \alpha_0, \quad X_1 \sim \alpha_1,$$

possibly with additional noise Z. Its conditional velocity is $u_t = \partial_t I_t(X_0, X_1, Z)$.

The goal of flow matching is to learn a vector field v_t whose continuity equation reproduces the law of X_t.

Live panel: flow-matching trajectories

schedule

vp

particles

60

60

atom gap

1.18

1,18

seed

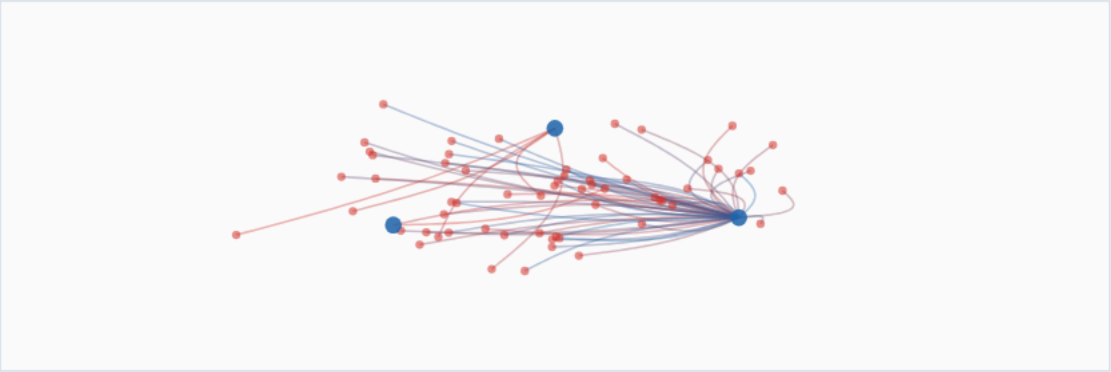
3704

3704

Reset

Open larger

vp reverse flow

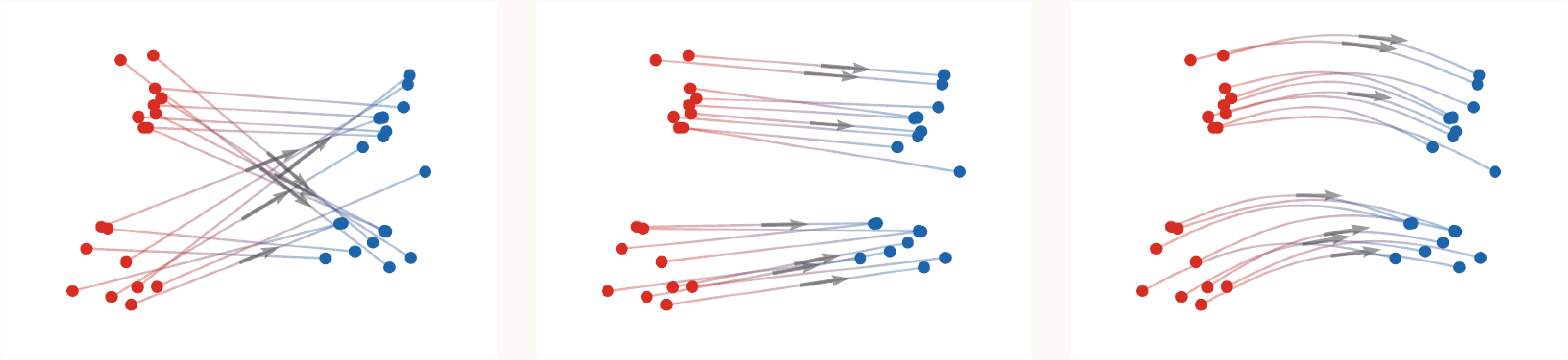


quadratic OT rays

vp schedule; 60 noise samples; atom gap 1.18; compare curved reverse flow with straight OT rays

Different endpoint couplings and schedules generate different training velocities.

Three interpolants



The coupling between endpoints and the time schedule change the geometry of the learned flow.

Flow matching objective

Let u_t be the conditional velocity of the interpolant. A common objective is

$$\min_v \int_0^1 \mathbb{E}\left[\|v_t(X_t) - u_t\|^2\right] dt.$$

Flow matching identity.
The minimizer is the conditional expectation

$$v_t(x) = \mathbb{E}[u_t \mid X_t = x],$$

and it transports the marginal law of X_t through the continuity equation.

Algorithm: flow matching training

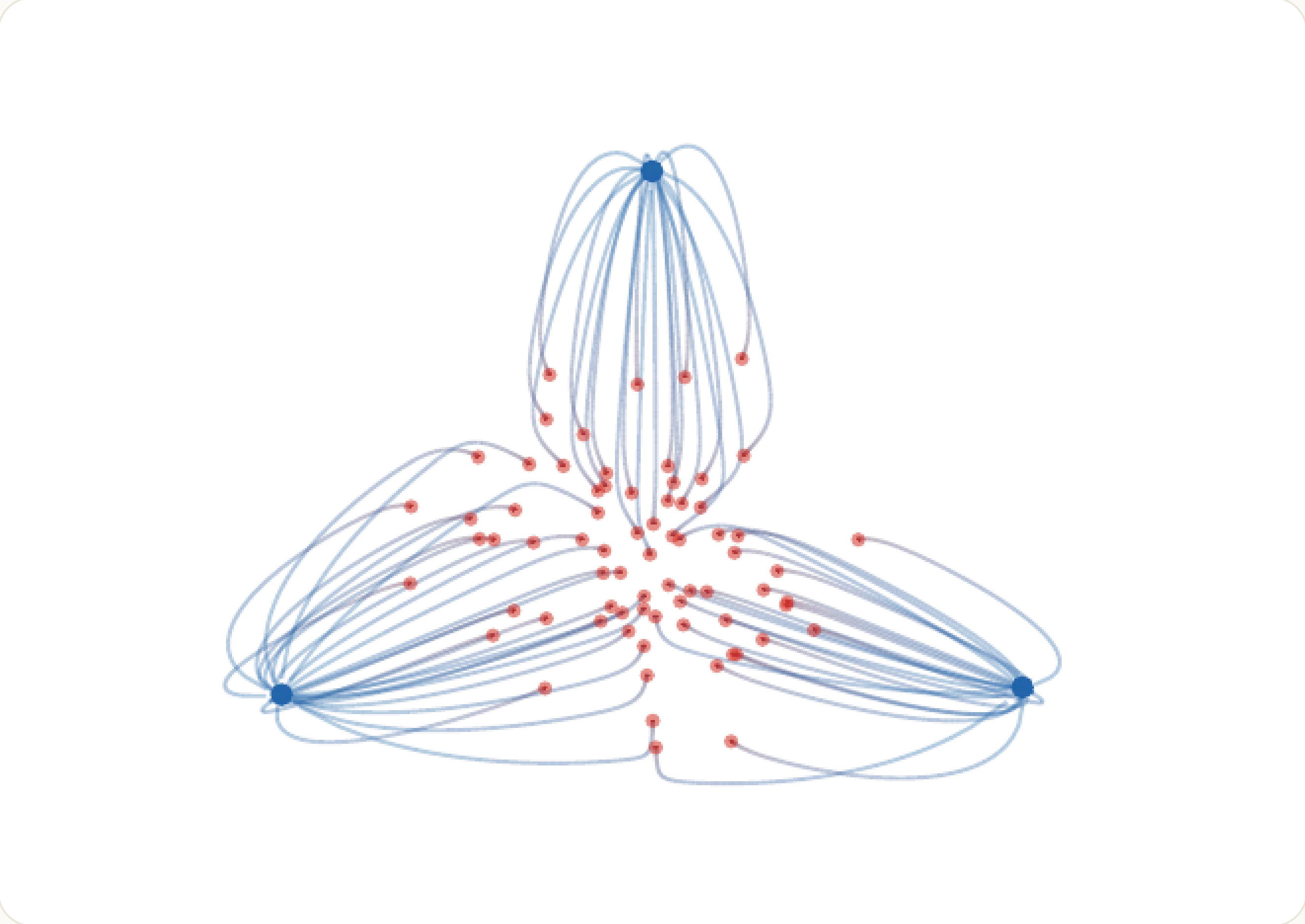
Flow matching from paired endpoint samples.

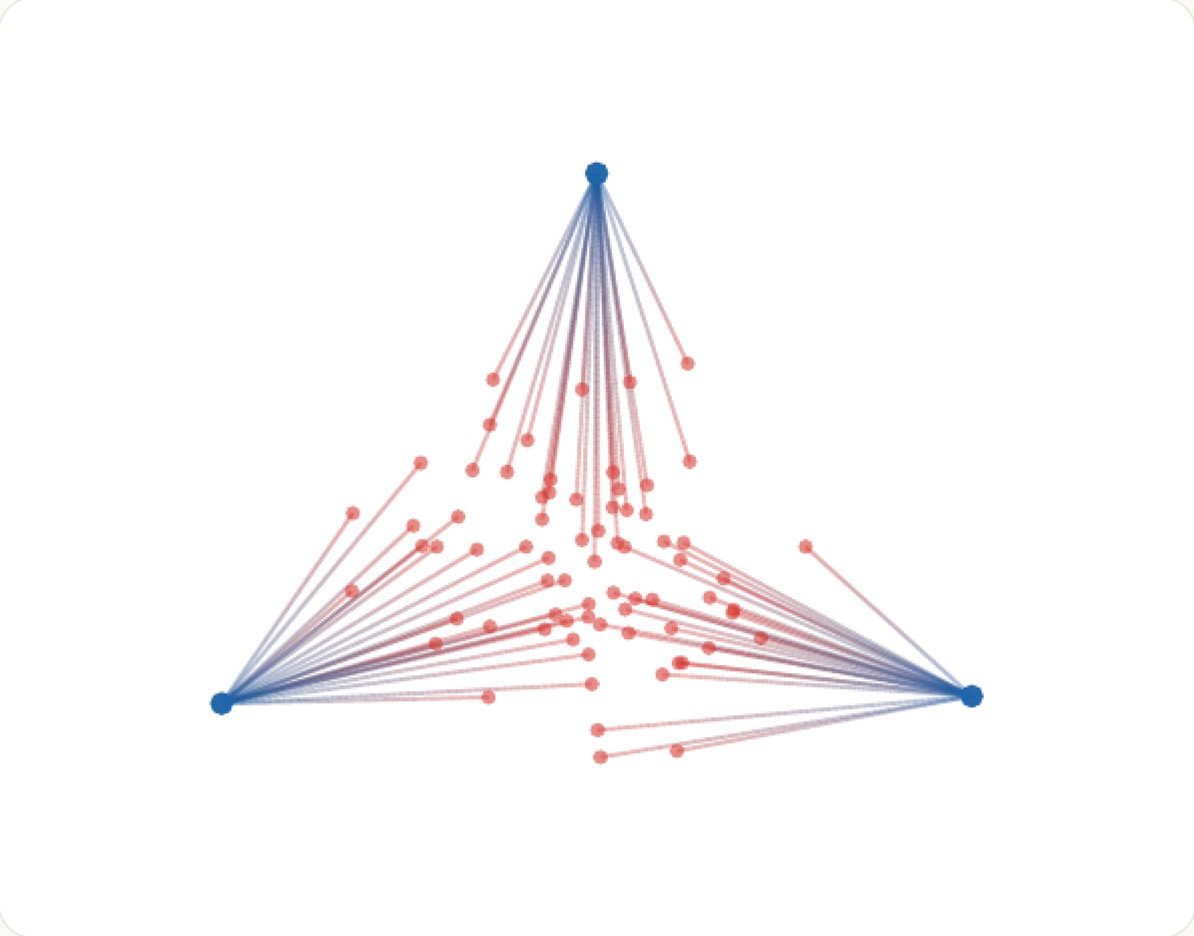
Input: endpoint sampler (X_0, X_1), schedule I_t, conditional velocity u_t, network $v_\theta(t, x)$.

Output: trained vector field v_θ.

1. Sample a mini-batch $(x_0^r, x_1^r)_r$.
2. Sample times $t_r \sim \mathrm{U}(0,1)$.
3. Form $x_t^r = I_{t_r}(x_0^r, x_1^r)$ and $u_t^r = u_{t_r}(x_0^r, x_1^r)$.
4. Minimize $\sum_r \|v_\theta(t_r, x_t^r) - u_t^r\|^2$ by stochastic gradient descent.
5. Generate samples by integrating $\dot{X}_t = v_\theta(t, X_t)$ from $X_0 \sim \alpha_0$.

Diffusion versus optimal transport





Diffusion-style paths can be more curved and noisy; OT paths are displacement-efficient but may be harder to parametrize in high dimension.

Live panel: diffusion and OT sampling

time
0.58
0,58

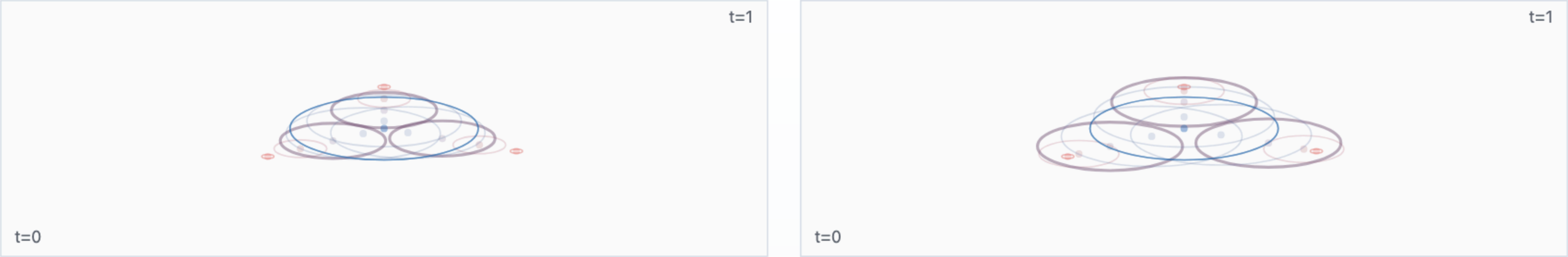
atom gap
1.15
1,15

noise
0.54
0,54

Reset Open larger

linear bridge, t=0.58

OU / variance-preserving bridge



selected time 0.58; atom gap 1.15; Gaussian endpoint sigma 0.54

Diffusion trajectories and OT trajectories are compared on the same target geometry.

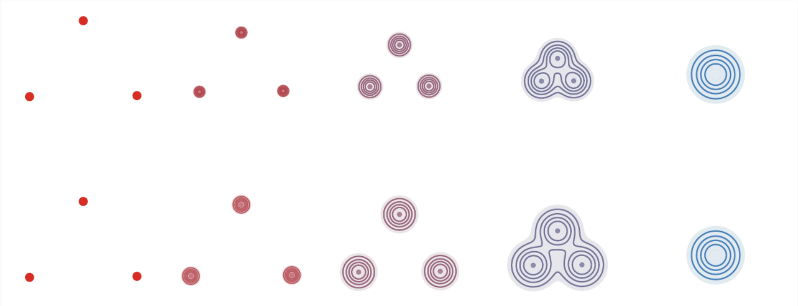
Is a diffusion map optimal?

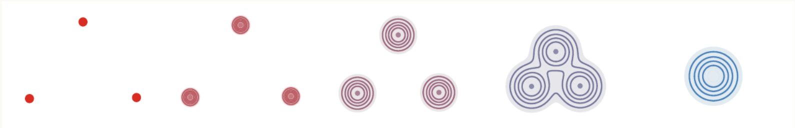
This question separates flow matching from OT geodesics.

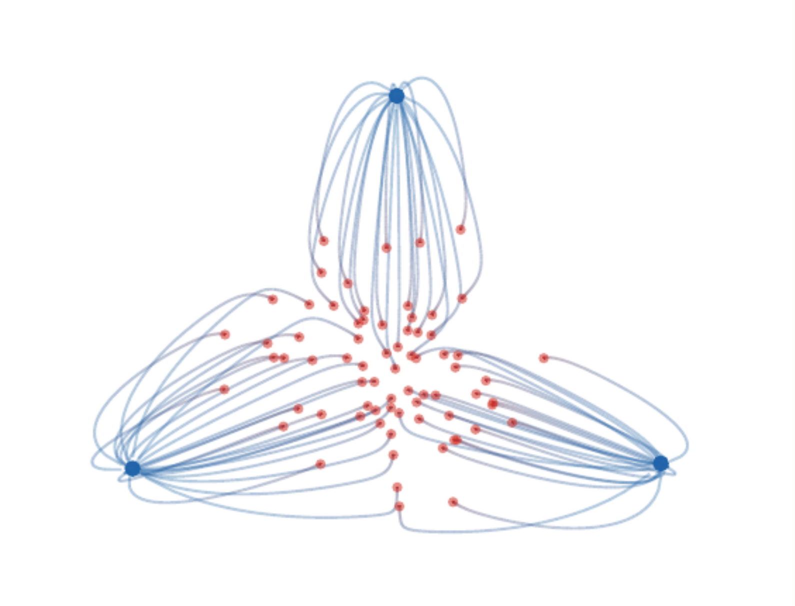
Key distinction.
Both diffusion-style interpolants and OT interpolants connect simple and complex laws, but only the Brenier interpolation is guaranteed to minimize kinetic action. A learned sampler can therefore be a valid transport without being an optimal transport map.

Changing the interpolant changes trajectory geometry even when the endpoints are fixed.

Forward noising schedules







Different schedules produce different velocity fields even when they connect similar endpoint distributions.

Live panel: noising schedules

particles
54

54

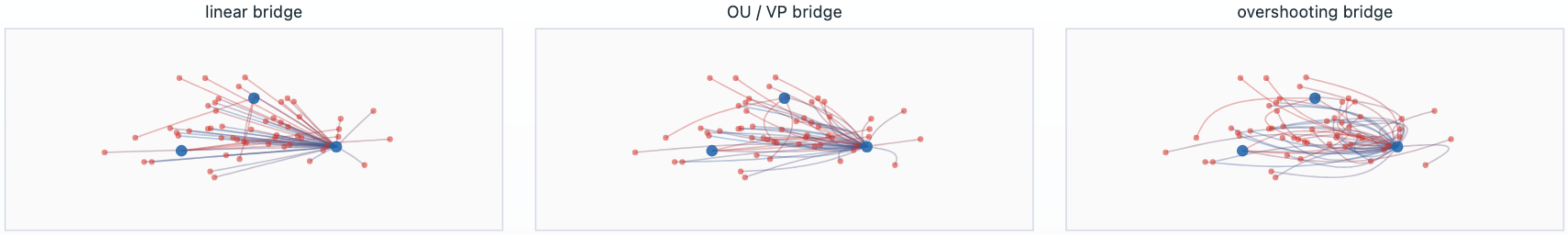
atom gap
1.18

1,18

seed
3742

3742

Reset
Open larger



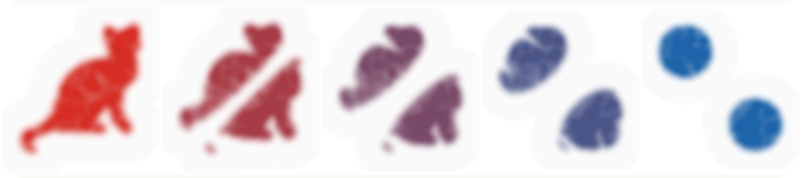
54 reverse paths; atom gap 1.18; the three scalar schedules share endpoints but bend the flow differently

The schedule changes the speed at which the same endpoint problem is traversed.

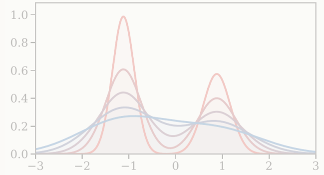
Roadmap: dynamic optimal transport

1. Generative transport

2. Dynamic OT



3. Gradient flows



4. Training and biology

5. Attention dynamics

Wasserstein distance revisited

For quadratic cost,

$$W_2^2(\alpha_0, \alpha_1) = \min_{\pi \in \Gamma(\alpha_0, \alpha_1)} \int \|x - y\|^2 \, d\pi(x, y).$$

In the Monge case, $\alpha_t = ((1-t)\,\mathrm{Id} + tT)_\# \alpha_0$ is a constant-speed geodesic.

Dynamic OT asks for an intrinsic description of this geodesic through velocities.

Benamou-Brenier formulation

Benamou-Brenier.
For absolutely continuous curves,

$$\frac{1}{2}W_2^2(\alpha_0, \alpha_1) = \min_{\rho_t, v_t} \int_0^1 \int \frac{1}{2}\|v_t(x)\|^2 \rho_t(x)\, dx\, dt$$

subject to

$$\partial_t \rho_t + \mathrm{div}(\rho_t v_t) = 0, \qquad \rho_{t=0} = \rho_0, \quad \rho_{t=1} = \rho_1.$$

Live panel: Benamou-Brenier geodesics

time
0.50
0,50

particles
42
42

velocity scale
0.90
0,90

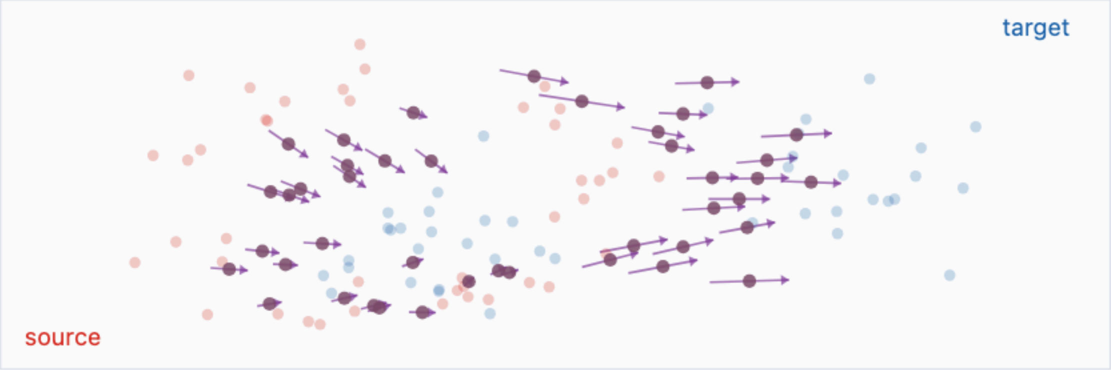
target
two blobs

seed
2710
2710

Reset
Open larger

McCann interpolation, t=0.50

Eulerian velocity samples



t=0.50; 42 matched characteristics; mean squared displacement 0.575; velocity scale 0.90

The dynamic formulation is easiest to read as mass moving through time.

Convex moment formulation

Set $m_t = \rho_t v_t$. Then the action becomes the perspective function

$$\int_0^1 \int \frac{1}{2} \frac{\|m_t(x)\|^2}{\rho_t(x)} \, dx \, dt,$$

with the linear constraint

$$\partial_t \rho_t + \mathbf{div}(m_t) = 0.$$

This is convex because $(\rho, m) \mapsto \|m\|^2/\rho$ is the perspective of a quadratic function.

Algorithm: dynamic OT solve

Convex Benamou-Brenier discretization.

Input: endpoint densities ρ_0, ρ_1, space-time grid, time step Δt.

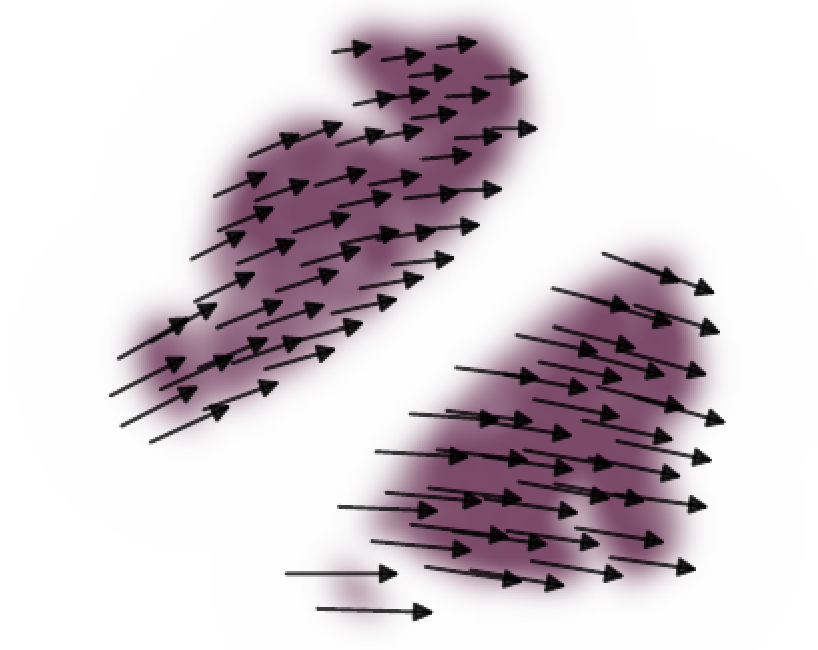
Output: density path $(\rho^k)_k$ and momenta $(m^k)_k$.

1. Initialize a feasible interpolation $(\rho^k, m^k)_k$.
2. Enforce the linear constraint $\partial_t \rho + \operatorname{div} m = 0$.
3. Minimize the convex action $\sum_k \Delta t \int \frac{1}{2} \|m^k\|^2 / \rho^k$.
4. Recover velocities by $v^k = m^k / \rho^k$ where $\rho^k > 0$.
5. Read the geodesic as the curve $\alpha_k = \rho^k dx$.

Dynamic geodesic example

The geodesic can be visualized either as evolving density or as a midpoint velocity field.

Velocity field



The vector field is not an arbitrary motion. It is the velocity minimizing kinetic energy under mass conservation.

At optimality, it is a gradient field in Euclidean quadratic transport.

Dual dynamic formulation

Hamilton-Jacobi dual.
The dual of the half-action formulation above can be expressed through potentials φ_t satisfying

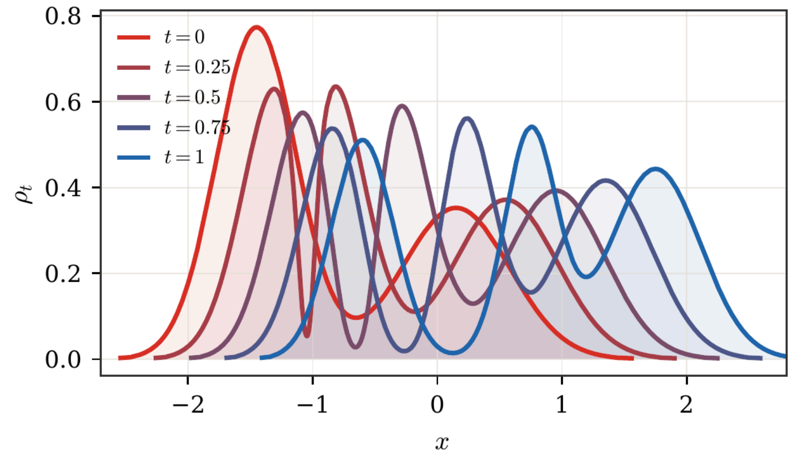
$$\partial_t \varphi_t + \frac{1}{2}\|\nabla \varphi_t\|^2 \leq 0.$$

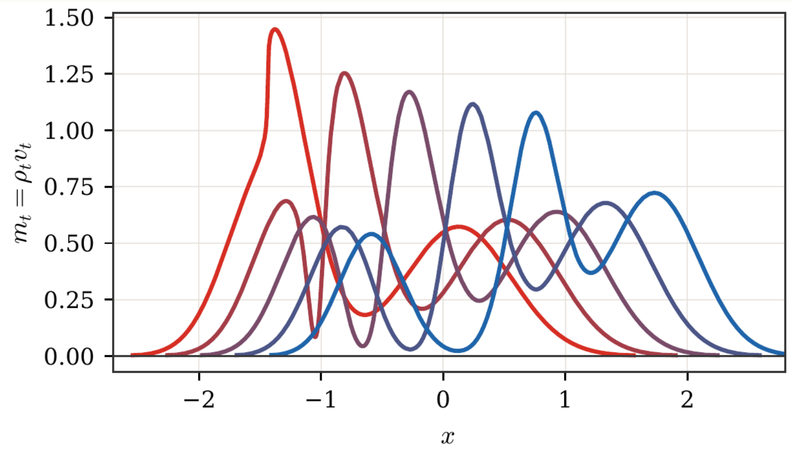
The endpoint objective is

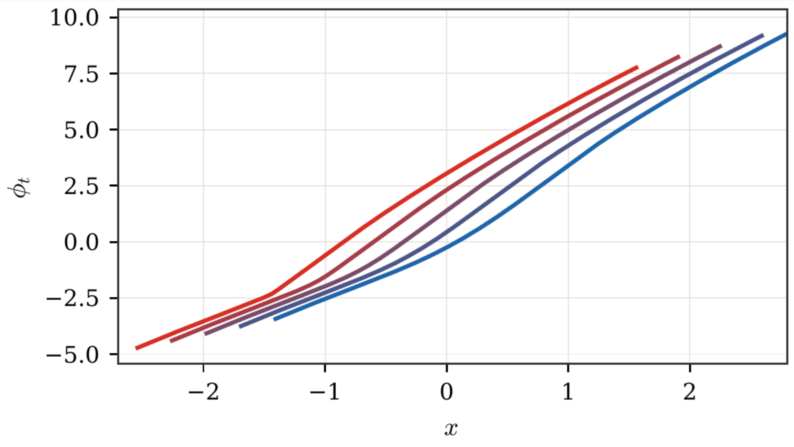
$$\int \varphi_1 \, d\alpha_1 - \int \varphi_0 \, d\alpha_0.$$

This is the dynamic counterpart of Kantorovich duality.

Primal and dual fields







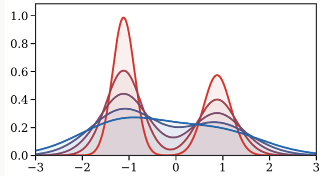
Density, momentum, and potential are coupled by the continuity equation and Hamilton-Jacobi inequality.

Roadmap: Wasserstein gradient flows

1. Generative transport

2. Dynamic OT

3. Gradient flows



4. Training and biology

5. Attention dynamics

Gradient flows in Wasserstein space

JKO scheme.
Given an energy $\mathcal{F}$, the implicit Euler step is

$$\alpha^{k+1} \in \arg\min_{\alpha} \frac{1}{2\tau} W_2^2(\alpha, \alpha^k) + \mathcal{F}(\alpha).$$

As $\tau \to 0$, this defines a gradient flow over probability measures.

Algorithm: minimizing movements

JKO time stepping.

Input: initial law α^0, energy $\mathcal{F}$, step size τ, number of steps K.

Output: approximate gradient-flow path $(\alpha^k)_{k=0}^K$.

1. Set $k = 0$.
2. Solve $\alpha^{k+1} \in \arg\min_\alpha \frac{1}{2\tau} W_2^2(\alpha, \alpha^k) + \mathcal{F}(\alpha)$.
3. Set $k \leftarrow k + 1$.
4. Repeat until $k = K$.
5. Interpolate the sequence to approximate the continuous curve α_t.

From JKO to PDE

When the first variation $\delta\mathcal{F}/\delta\rho$ is smooth, the Wasserstein gradient flow satisfies

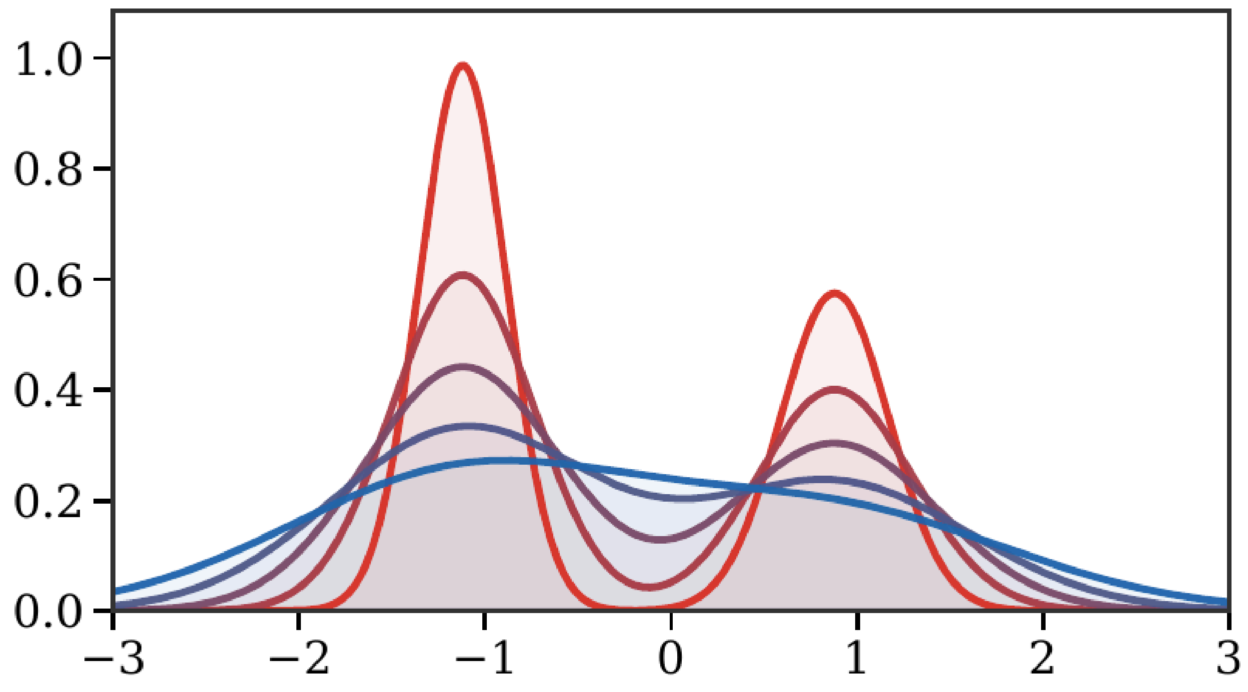
$$\partial_t \rho_t = \operatorname{div}\left(\rho_t \nabla \frac{\delta \mathcal{F}}{\delta \rho}(\rho_t)\right).$$

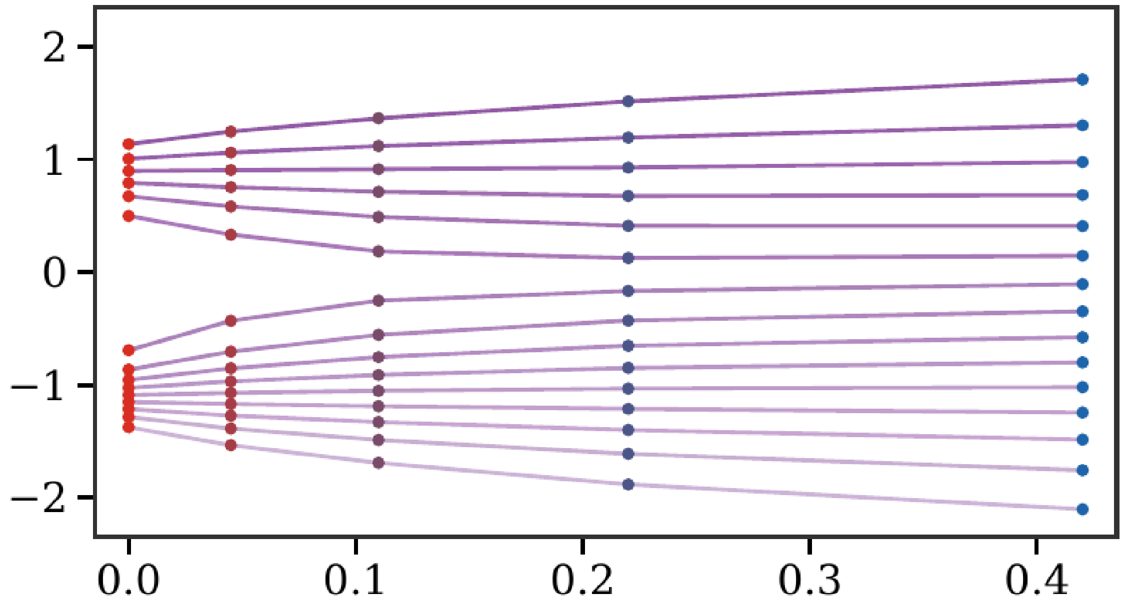
Equivalently,

$$v_t = -\nabla \frac{\delta \mathcal{F}}{\delta \rho}(\rho_t), \qquad \partial_t \rho_t + \operatorname{div}(\rho_t v_t) = 0.$$

For example, $\delta\, \mathrm{Ent}(\rho)/\delta\rho = \log \rho + 1$ gives the heat equation.

JKO entropy steps





The implicit step regularizes the density by moving in Wasserstein geometry rather than in a fixed Euclidean space.

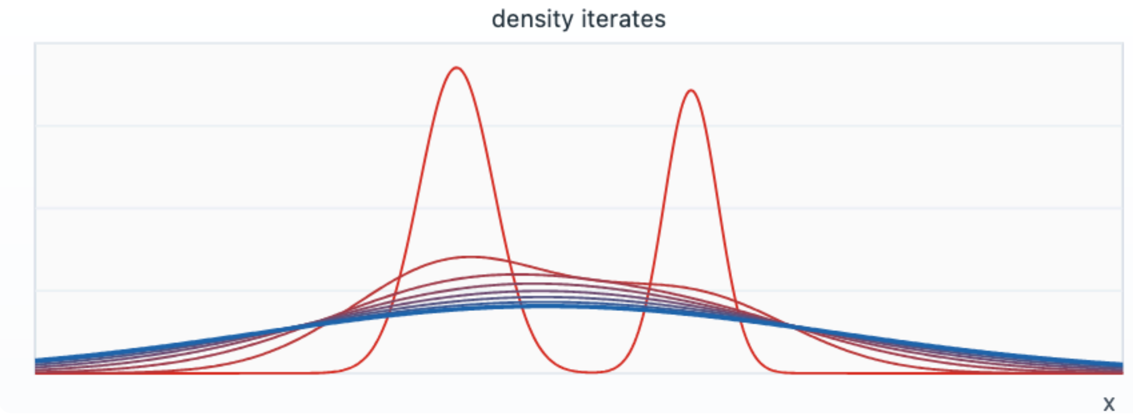
Live panel: JKO minimizing movements

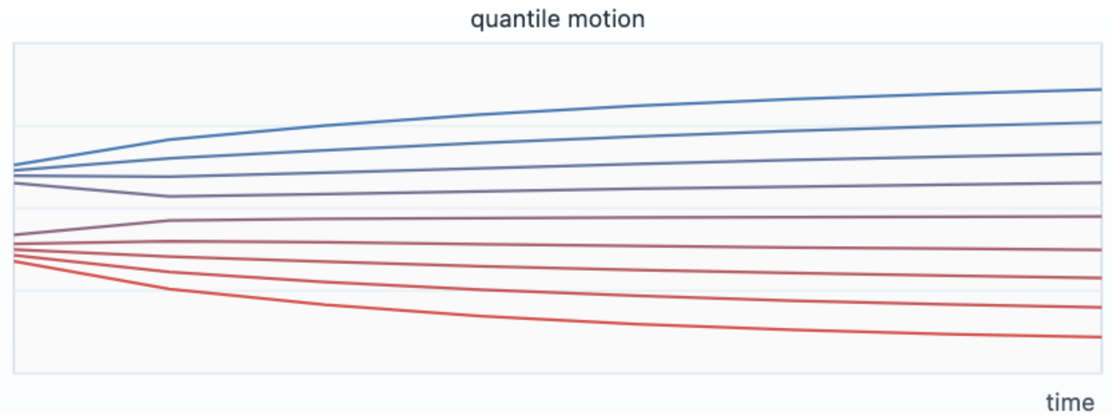
step size
0.16
0,16

steps
7
7

initial width
0.22
0,22

Reset
Open larger





7 JKO steps; tau=0.16; final heat time 1.12

The implicit Euler step trades decrease of energy against movement in Wasserstein space.

Gradient structure

Otto metric intuition.

The tangent vector $\partial_t \rho = -\operatorname{div}(\rho v)$ is measured by

$$\|\partial_t \rho\|_{T_\rho}^2 = \inf_v \left\{ \int \|v\|^2 d\rho : \partial_t \rho + \operatorname{div}(\rho v) = 0 \right\}.$$

This is the Riemannian-like structure behind W_2 gradient flows.

Least-squares inversion

The velocity is not arbitrary: for a prescribed infinitesimal density variation, the Wasserstein metric selects the minimum-energy vector field.

Otto inversion.

For $\dot{\rho}$ with zero mass,

$$\|\dot{\rho}\|_{T_\rho}^2 = \min_v \left\{ \int \|v(x)\|^2 \rho(x) dx : \dot{\rho} + \mathrm{div}(\rho v) = 0 \right\}.$$

The minimizer is a gradient field $v = \nabla\psi$, where

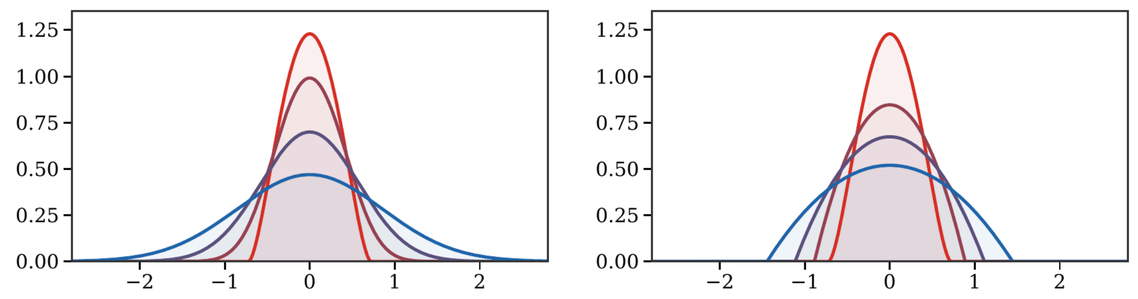
$$\Delta_\rho \psi = \mathrm{div}(\rho \nabla \psi) = -\dot{\rho}.$$

This elliptic inversion is the practical bottleneck behind many high-dimensional Wasserstein flows.

Linear and nonlinear diffusions

Entropy-type energies give diffusion PDEs:

- Shannon entropy gives the heat equation,
- power entropy gives porous-medium equations.



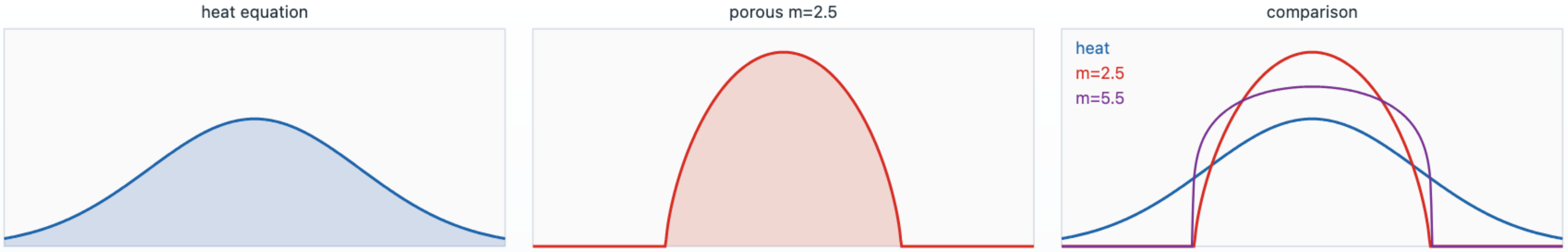
Live panel: diffusion gradient flows

time
0.72
0,72

power m
2.5
2,5

initial width
0.62
0,62

Reset | Open larger



time 0.72; porous exponent m=2.5; compact front radius grows slowly as m increases

Entropy and porous-medium energies produce different spreading laws.

Fokker-Planck and Langevin dynamics

For relative entropy with respect to a target density $\beta \propto e^{-V}$,

$$\mathcal{F}(\alpha) = \mathrm{KL}(\alpha \mid \beta),$$

the flow is the Fokker-Planck equation

$$\partial_t \rho = \mathrm{div}(\rho \nabla V) + \Delta \rho.$$

This is the law of Langevin particles

$$dX_t = -\nabla V(X_t)dt + \sqrt{2}\, dB_t.$$

Thus the same evolution can be read as a deterministic PDE for ρ_t or as the law of a stochastic particle system.

Live panel: Fokker-Planck evolution

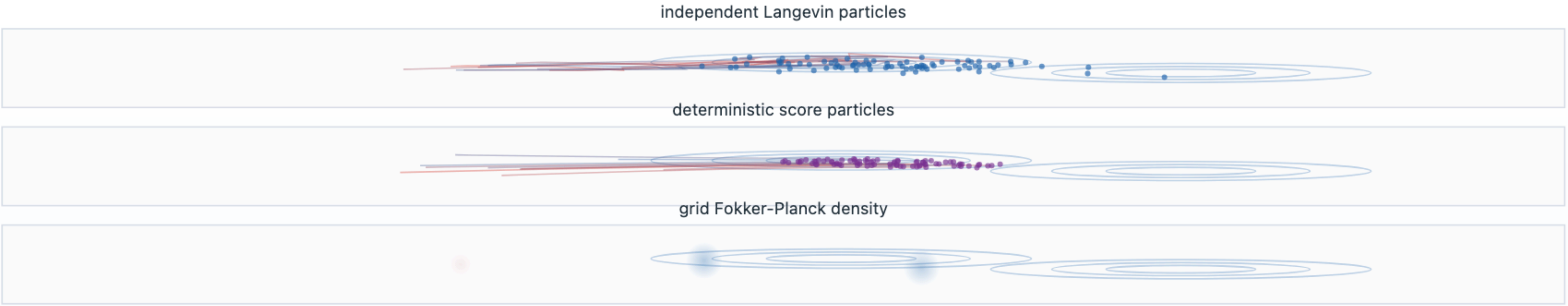
time
0.64
0,64

noise
0.34
0,34

target shift
0.62
0,62

seed
3408
3408

Reset
Open larger



time 0.64; noise sigma 0.34; target shift 0.62

The deterministic PDE and the particle SDE give two numerical views of the same flow.

Interactive: deterministic versus noisy descent

Noise changes the geometry of particle trajectories. With zero noise, particles follow descent curves. With Langevin noise, they explore the target landscape and spread across modes.

The slider controls the Brownian amplitude in a two-well potential.

Algorithm: Langevin particle solver

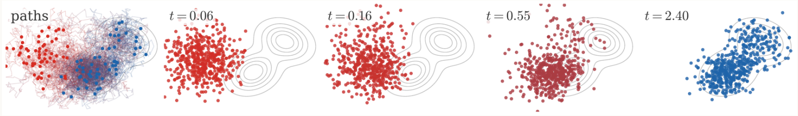
Euler-Maruyama Fokker-Planck sampler.

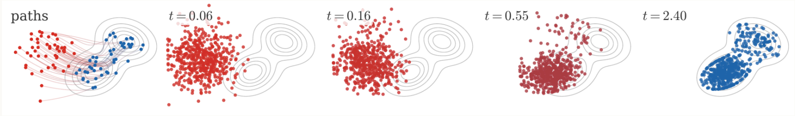
Input: target potential V, step size h, particles $(X_i^0)_{i=1}^N$.

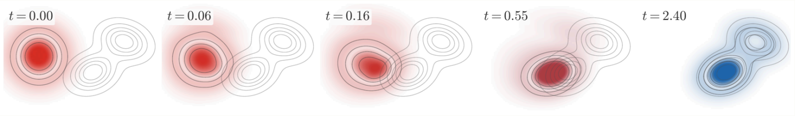
Output: empirical approximation $\alpha_t^N = \frac{1}{N}\sum_i \delta_{X_i^t}$.

1. For each step $k = 0, \dots, K-1$, sample $\xi_i^k \sim \mathrm{N}(0, I_d)$.
2. Update $X_i^{k+1} = X_i^k - h\nabla V(X_i^k) + \sqrt{2h}\,\xi_i^k$.
3. Approximate the law by $\alpha_{kh}^N = \frac{1}{N}\sum_i \delta_{X_i^k}$.
4. Estimate densities by KDE or solve the Fokker-Planck PDE directly when dimension permits.

Three numerical representations

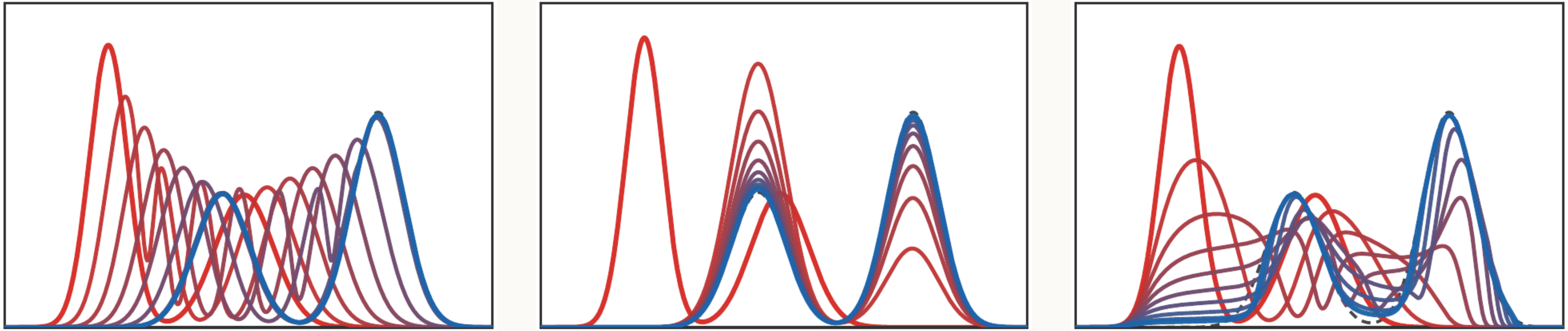






The same PDE can be approximated by stochastic particles, deterministic KDE-score particles, or a finite-difference density solver.

Discrepancy-driven flows



Changing the discrepancy changes the geometry of the velocity field and the resulting motion.

Live panel: gradient-flow objectives

geometry

sinkhorn

smoothing

0.55

0,55

particles

28

28

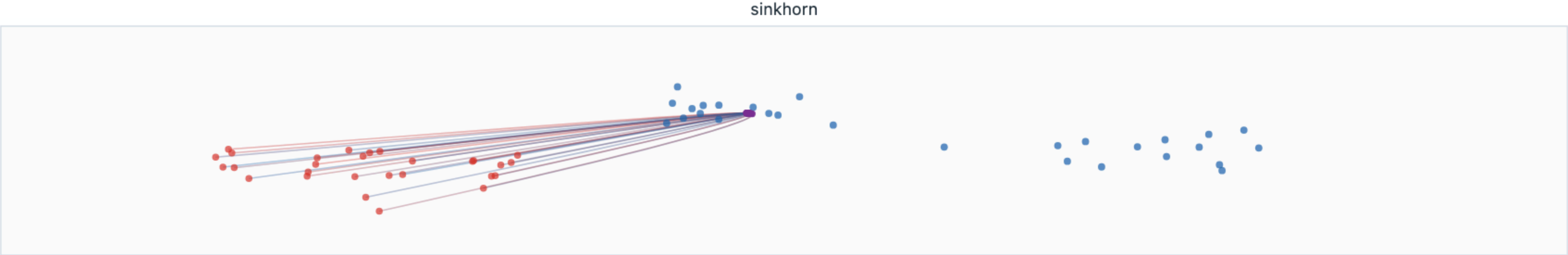
seed

3302

3302

Reset

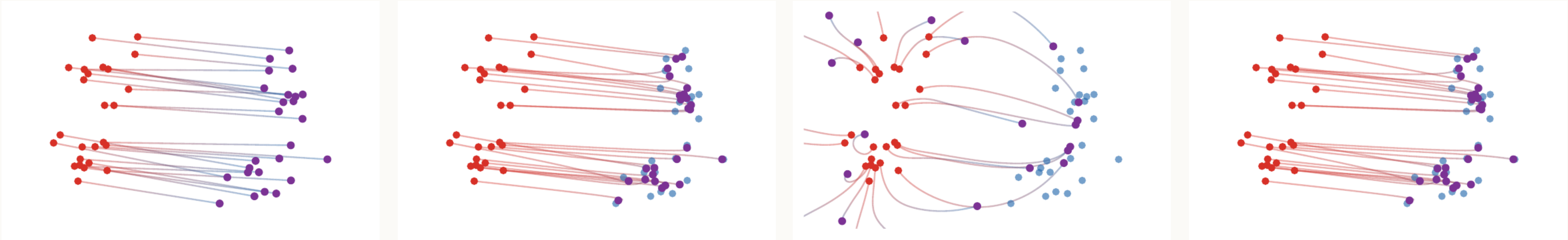
Open larger



sinkhorn geometry; smoothing 0.55; 28 source particles

Different discrepancies create different velocities even with the same target.

Particle objective geometries



Different losses induce very different particle forces: OT rays, local-density pressure, MMD fields, and Sinkhorn divergence forces.

Live panel: MMD particle flow

particles
64
64

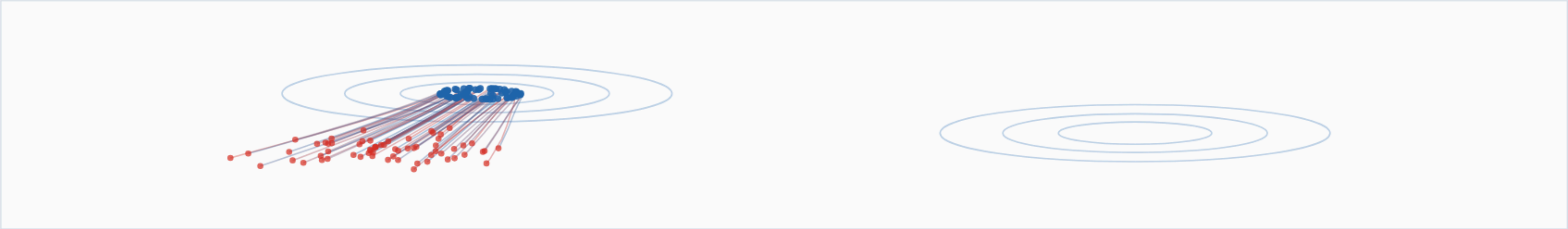
mode gap
2.10
2,10

kernel width
0.62
0,62

seed
3104
3104

Reset Open larger

deterministic MMD-type particle flow

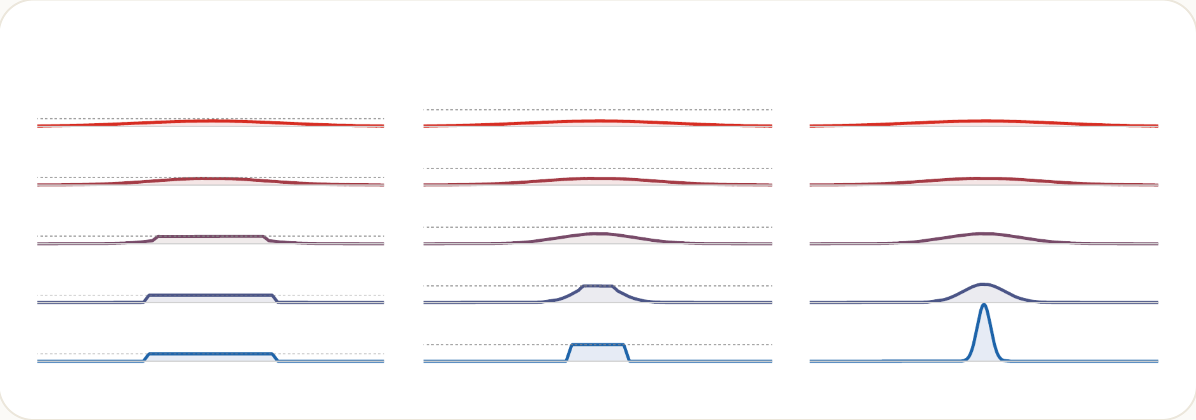


64 particles; kernel width 0.62; teacher separation 2.10; mean distance to nearest mode 0.10

The particle count changes how faithfully the flow explores the target geometry.

Constraints and unbalanced flows

Dynamic formulations can impose extra constraints or allow mass variation. These change the tangent space and therefore the PDE.



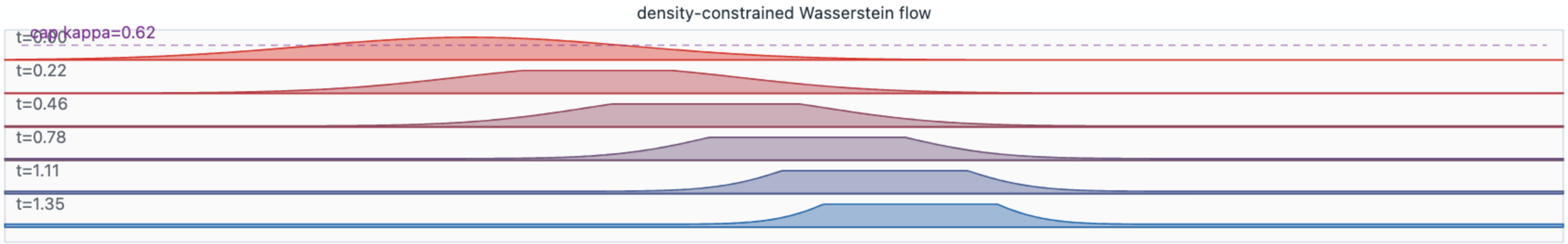
Live panel: density-constrained flows

density cap
0.62
0,62

attraction
1.25
1,25

final time
1.35
1,35

Reset Open larger



projected toy flow with density cap kappa=0.62; attraction 1.25; final time 1.35

Hard density caps create pressure-like effects in the transport dynamics.

Crowd motion with congestion

Congestion models combine transport, entropy, and a hard density cap. A typical energy is

$$\mathcal{F}(\alpha) = \int \varphi \, d\alpha + \mathrm{Ent}(\alpha) + \iota_C(\alpha), \qquad C = \{\rho \, dx : \rho(x) \leq \kappa(x)\}.$$

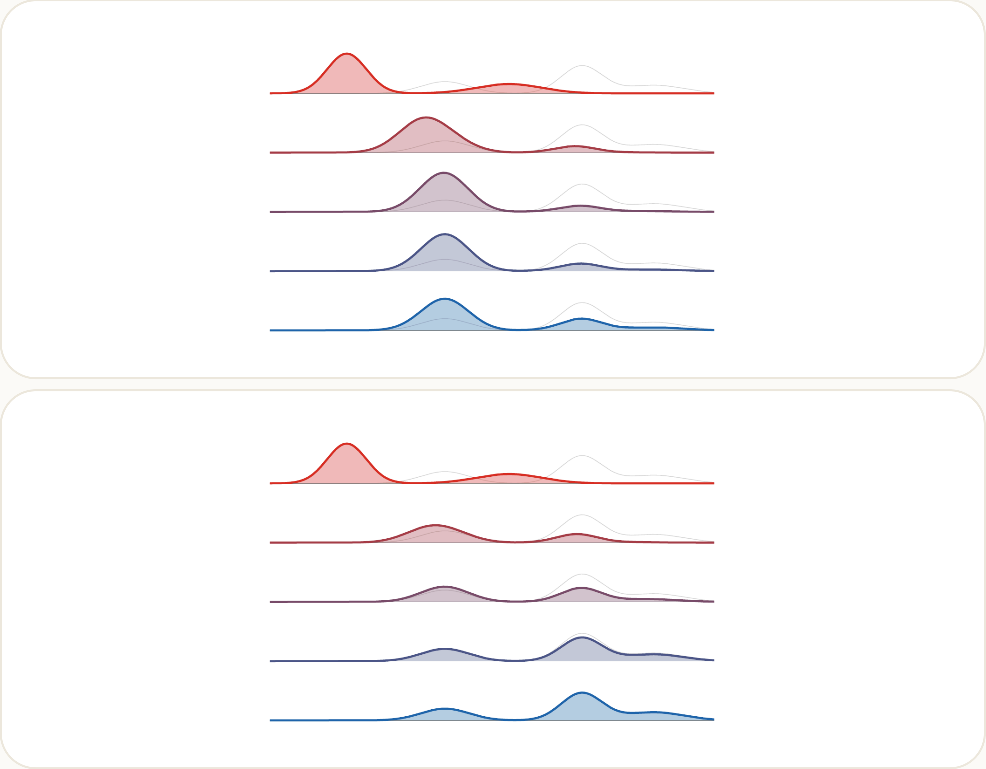
Example: density cap.
The Wasserstein gradient flow follows the force $-\nabla\varphi$, diffuses through entropy, and adds a pressure term enforcing $\rho \leq \kappa$.

The reference model is the Maury-Roudneff-Chupin-Santambrogio crowd-motion framework.

Crowd motion visual

This visual illustrates how the hard density cap creates a pressure field in congested regions.

Wasserstein-Fisher-Rao flows

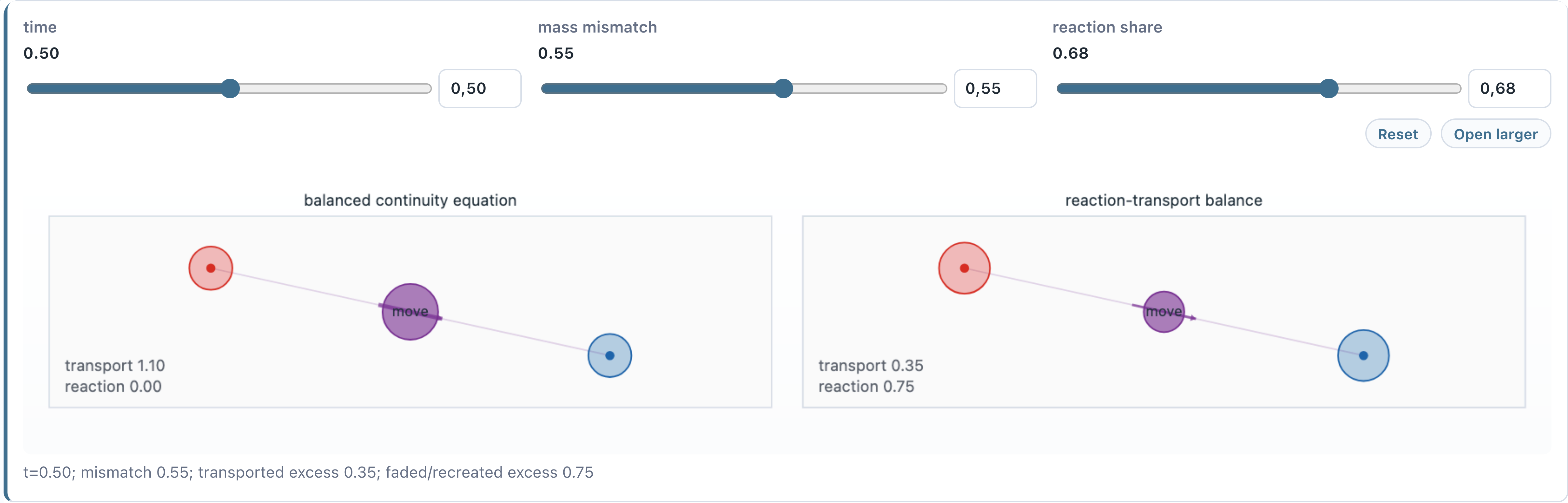


WFR augments transport by growth:

$$\partial_t \rho + \mathrm{div}(\rho v) = \rho r, \qquad \int_0^1 \int (\|v\|^2 + r^2)\rho \, dx \, dt.$$

Unbalanced geometries add a scalar growth field to the transport velocity, allowing mass to appear or disappear.

Live panel: dynamic unbalanced flows



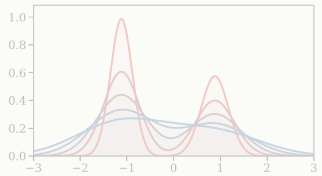
Unbalanced flows add growth and decay to pure transport.

Roadmap: training and biology

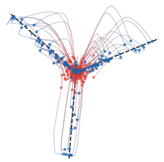
1. Generative transport

2. Dynamic OT

3. Gradient flows



4. Training and biology



5. Attention dynamics

Mean-field two-layer networks

A two-layer model can be written as an integral over neurons:

$$f_\alpha(z) = \int a\,\sigma(w^\top z)\,d\alpha(a, w).$$

Here α is a probability measure over neuron parameters (a, w). In the infinite-width limit, particle descent becomes a Wasserstein gradient flow for α_t.

Mean-field risk and first variation

Write $\theta = (a, w)$ and $\varphi_\theta(z) = a\,\sigma(w^\top z)$. For supervised data law ν on pairs (z, y),

$$\mathcal{R}(\alpha) = \frac{1}{2}\int |f_\alpha(z) - y|^2\, d\nu(z, y), \qquad f_\alpha(z) = \int \varphi_\theta(z)\, d\alpha(\theta).$$

The first variation is explicit:

$$\frac{\delta \mathcal{R}}{\delta \alpha}(\theta) = \int (f_\alpha(z) - y)\, \varphi_\theta(z)\, d\nu(z, y).$$

Particle gradient descent differentiates this quantity with respect to the neuron parameter θ.

Live panel: mean-field MLP training

neurons
90
90

ray angle
34
34

time
0.72
0,72

seed
3506
3506

Reset Open larger

90 neurons; teacher rays at +/- 34 degrees; displayed time 0.72

The neuron distribution evolves as a Wasserstein gradient flow in parameter space.

Mean-field training PDE

For a risk functional $\mathcal{R}(\alpha)$, the mean-field dynamics has the form

$$\partial_t \alpha_t + \operatorname{div}(\alpha_t v_t) = 0, \qquad v_t = -\nabla \frac{\delta \mathcal{R}}{\delta \alpha}(\alpha_t).$$

This is a training PDE, not merely an optimization algorithm in a finite parameter vector.

Perceptron global-convergence message

For two-layer perceptrons, the empirical particle system is

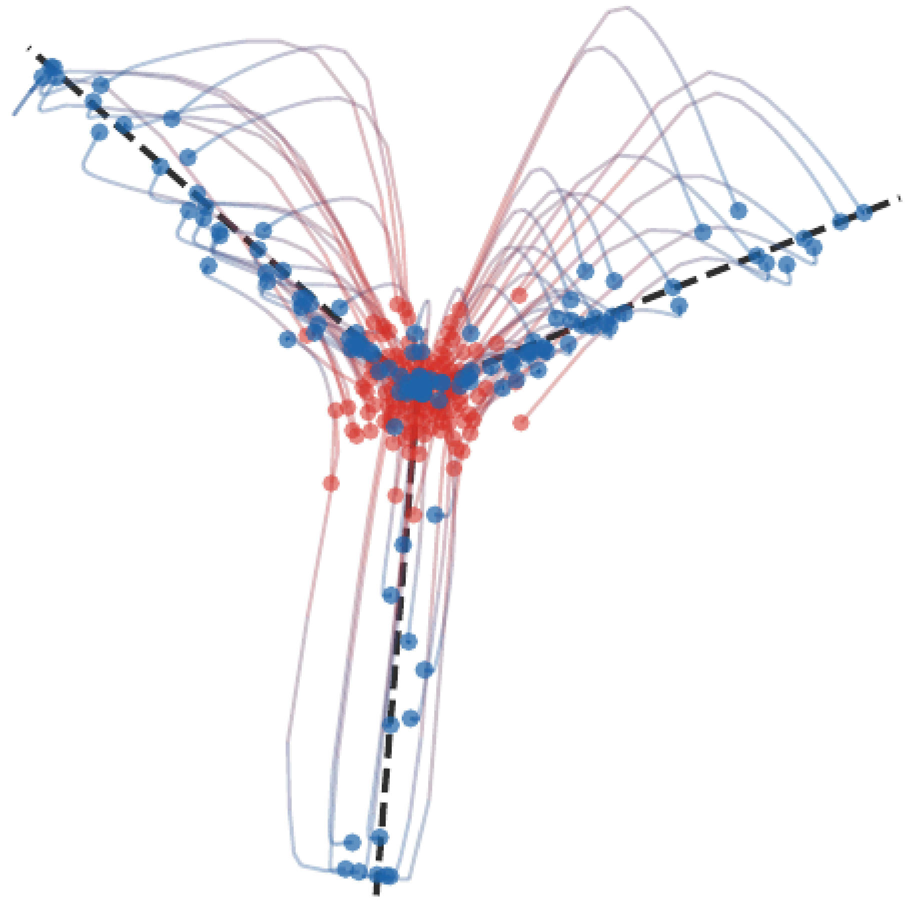
$$\alpha_t = \frac{1}{m}\sum_{k=1}^{m} \delta_{(a_k(t),w_k(t))}.$$

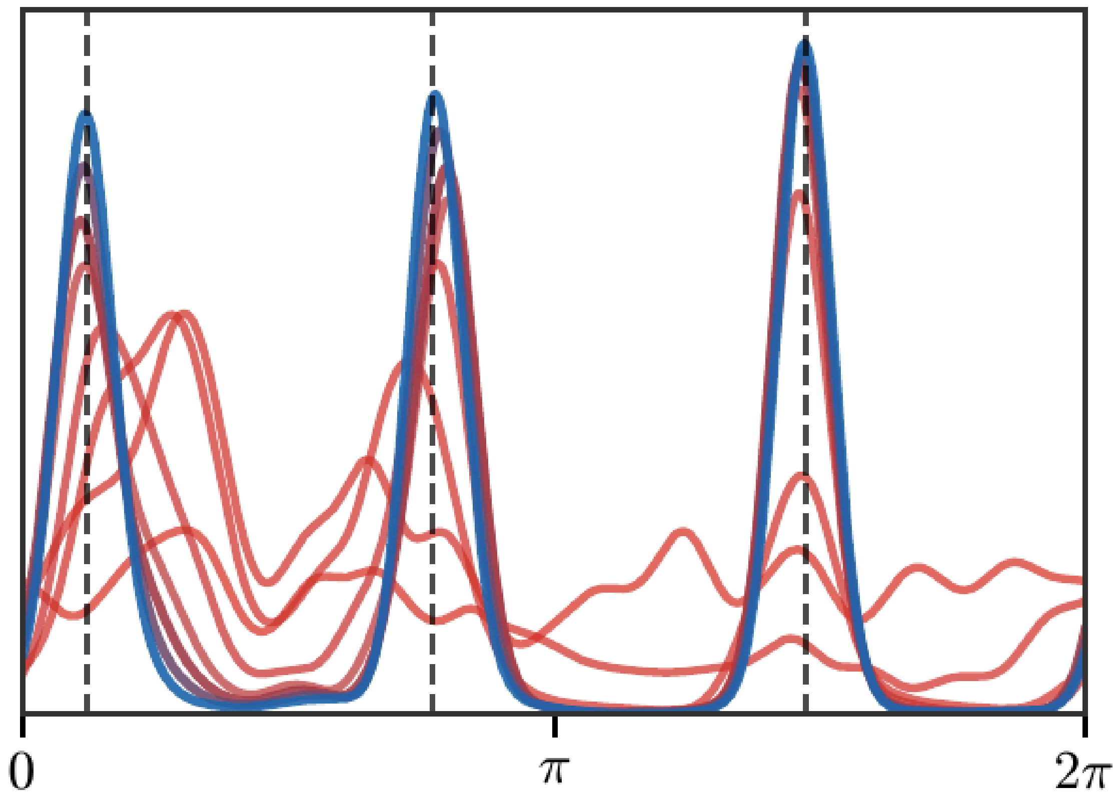
The mean-field limit replaces finite-dimensional parameter descent by descent of a functional $\mathcal{R}(\alpha)$.

Qualitative theorem.
With enough directional support, convergence rules out bad stationary points: the limiting measure is a global minimizer of the mean-field risk.

This is why the measure viewpoint is not just notation: it changes the convexity and stationarity picture.

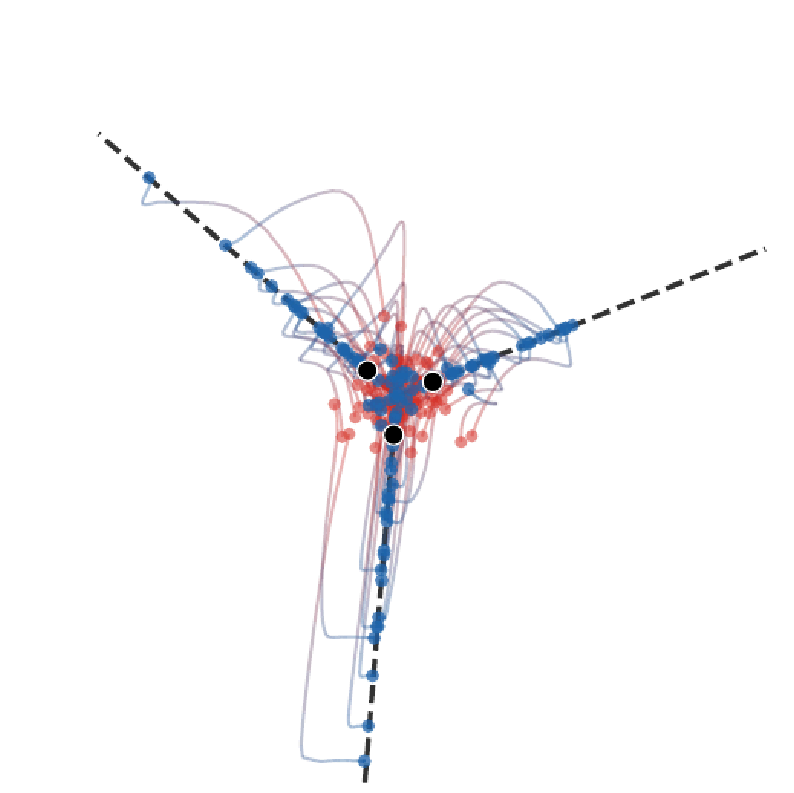
Neuron trajectories

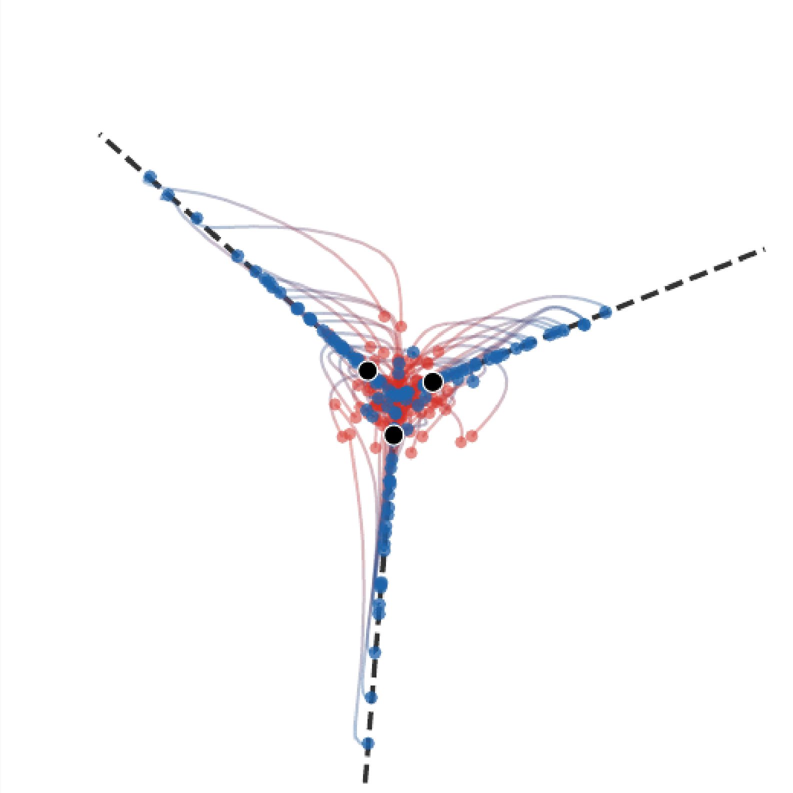


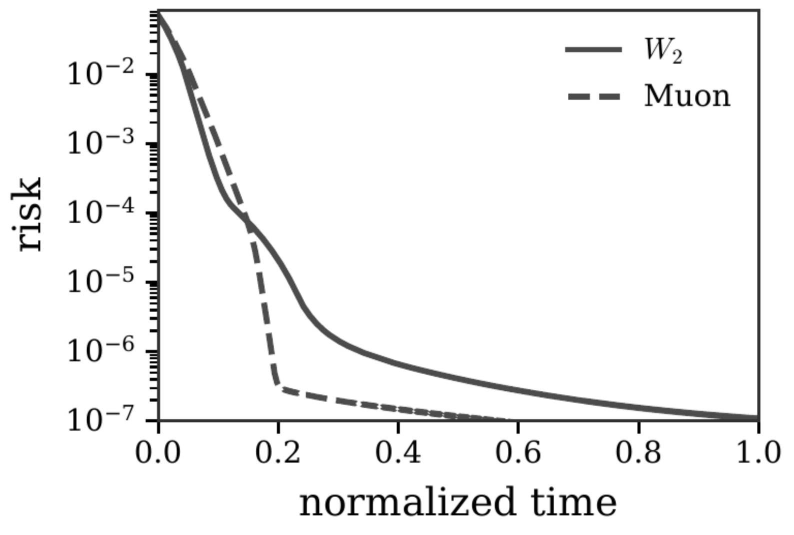


The empirical measure of neurons evolves as particles in parameter space.

Normalized and Muon-type flows







Changing the underlying transport geometry changes the training path and the risk decay.

Live panel: Muon-type normalized dynamics

neurons
92
92

teacher angle
38
38

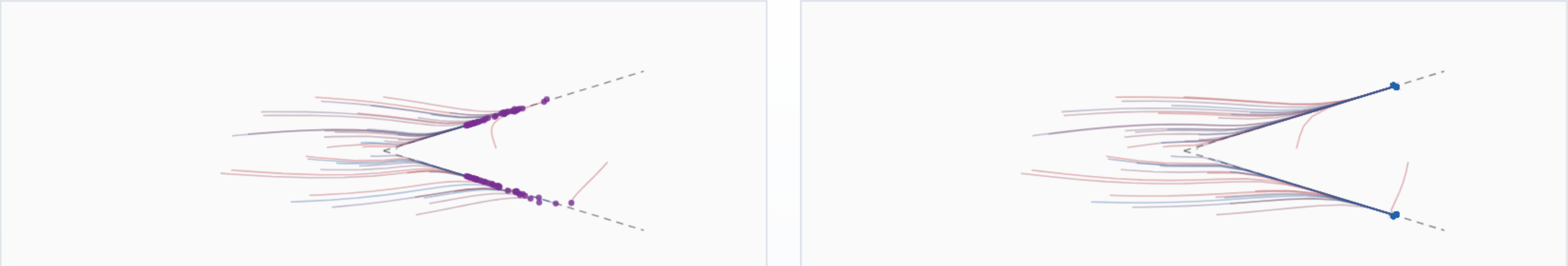
steps
86
86

seed
4830
4830

Reset
Open larger

W2 particle flow

normalized Muon-like flow



surrogate normalized training dynamics; final risk W2=0.018, Muon=0.000.

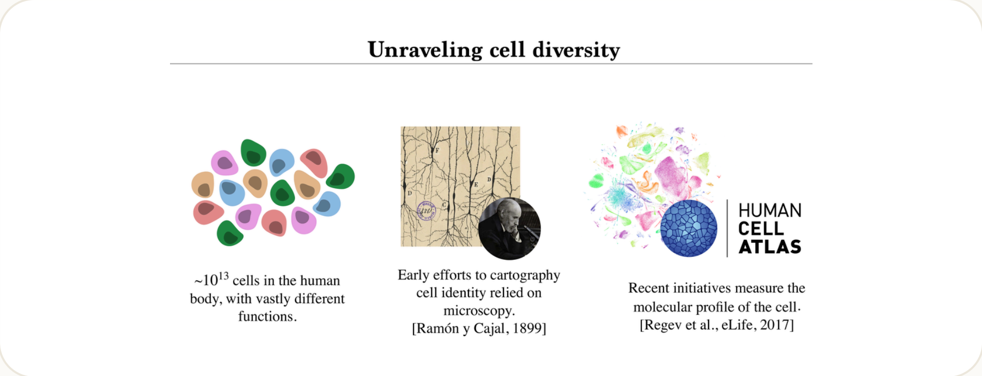
The normalized geometry changes the effective path through parameter space.

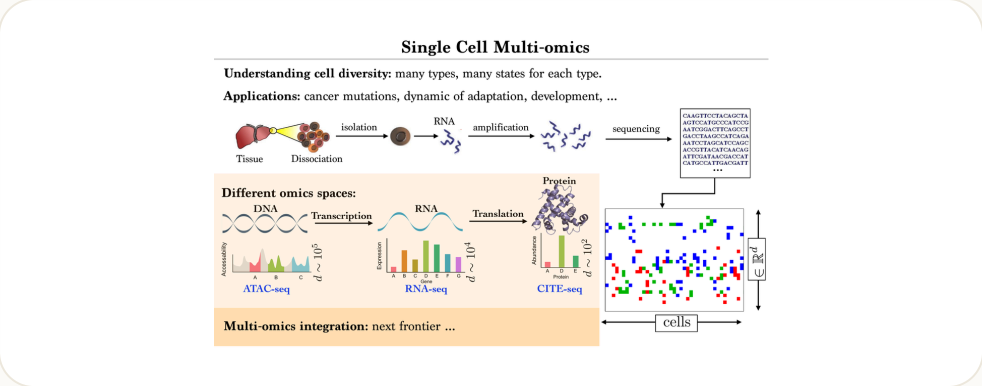
Single-cell and multi-omics viewpoint

Single-cell data naturally produces empirical distributions of cells in feature spaces. OT is used to:

- compare cell populations,
- match cells across modalities,
- infer developmental trajectories,
- regularize couplings with biological priors.

The mathematical object is again a coupling or a flow between distributions.





Cell-to-cell dissimilarity

Single-cell comparison is naturally a distribution problem rather than a pointwise one.

Example: cell populations.
A sample is an empirical measure

$$\alpha = \frac{1}{n}\sum_i \delta_{x_i}$$

over cells represented by gene-expression, accessibility, or learned latent features. OT compares two populations by coupling cells while respecting the geometry of the feature space.

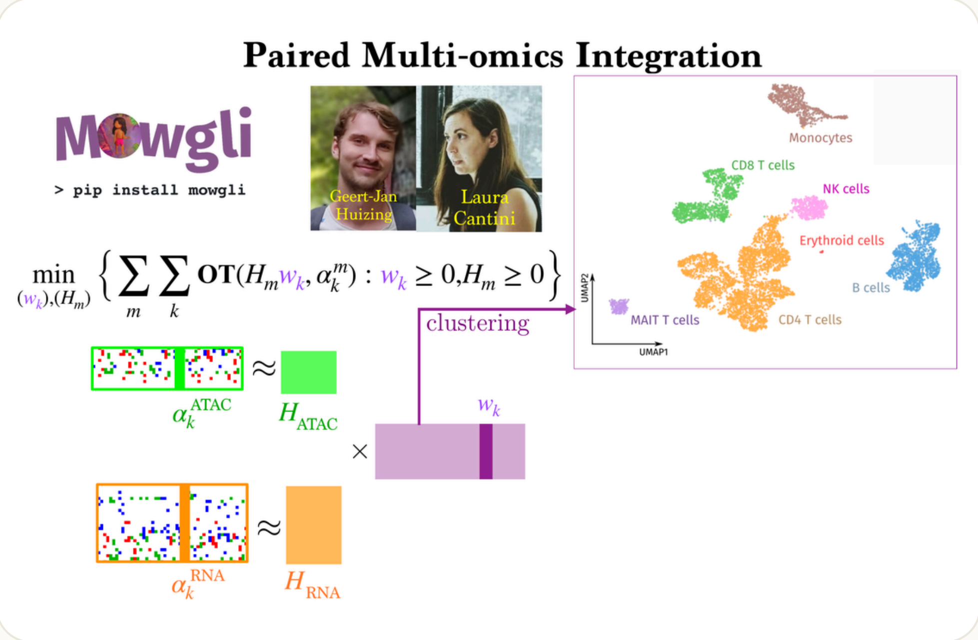
The coupling is not only a number: it can be interpreted as a soft cell-to-cell correspondence.

Paired multi-omics integration

When several modalities describe the same biological system, OT can align their empirical laws through a shared latent geometry.

Coupling across modalities.
Multi-omics integration often seeks nonnegative weights or maps that match several observed distributions at once. OT supplies the cost-sensitive alignment term, while additional constraints encode marker structure, sparsity, or temporal consistency.

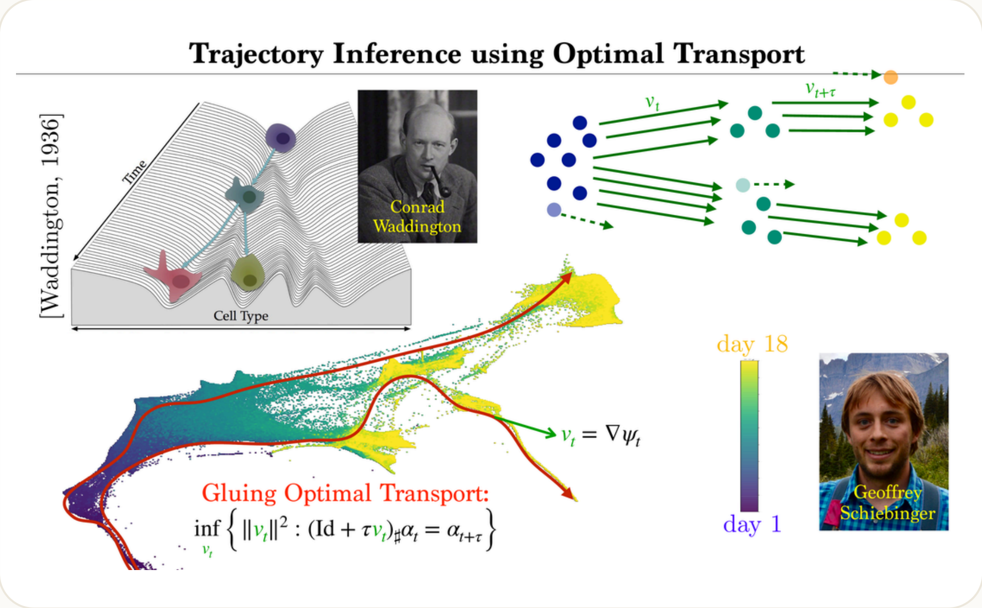
This is the same mathematical pattern as barycenters and multimarginal couplings, specialized to biological measurements. The MOWGLI model is a representative multi-omics example: it combines cost-sensitive OT couplings with low-rank factors to align modalities.



Trajectory inference as transport

Given observations at several times, one can infer a coupling or flow linking consecutive distributions.

Example: developmental dynamics.
A population at time t_k is a measure α_k. OT-based trajectory inference connects α_k to α_{k+1} with geometry and growth priors.

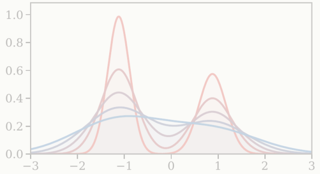


Roadmap: attention dynamics

1. Generative transport

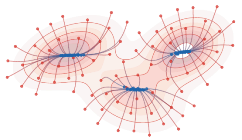
2. Dynamic OT

3. Gradient flows



4. Training and biology

5. Attention dynamics



Attention as interacting particles

Self-attention maps a set of tokens $(x_i)_i$ to weighted averages

$$A_i(X) = \sum_j \frac{\exp(q(x_i)^\top k(x_j))}{\sum_\ell \exp(q(x_i)^\top k(x_\ell))} v(x_j).$$

The maps q, k, v are the query, key, and value parametrizations.

This is permutation-equivariant and can be interpreted as an interaction rule between particles.

Continuous-depth transformers

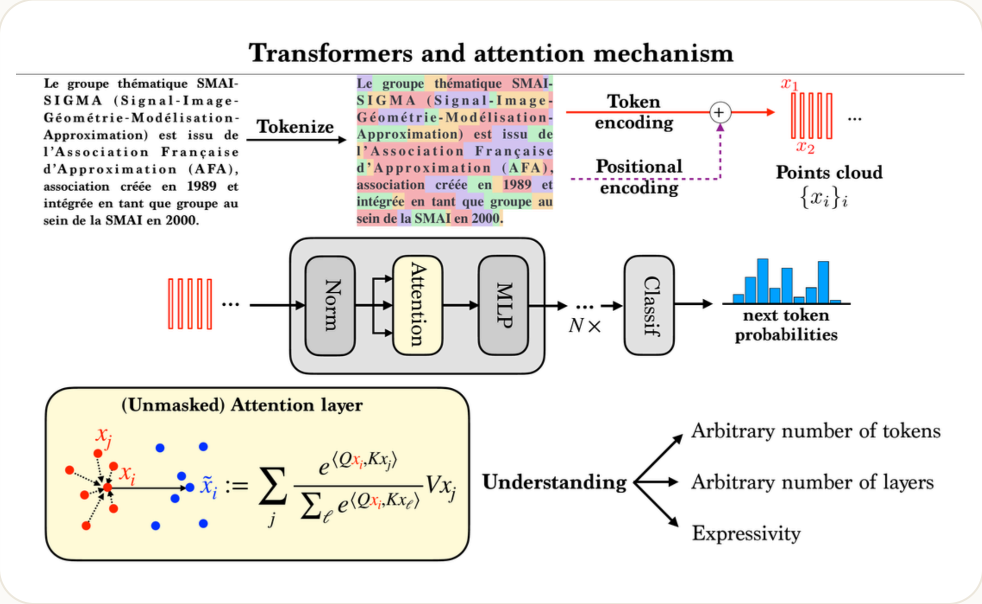
A residual transformer layer has the form

$$x_i^{\ell+1} = x_i^{\ell} + h\, A_i(X^{\ell}).$$

As the depth step $h \to 0$, this suggests an ODE

$$\frac{dx_i}{dt} = A_i(X_t),$$

and, in the mean-field limit, a PDE for the token distribution.



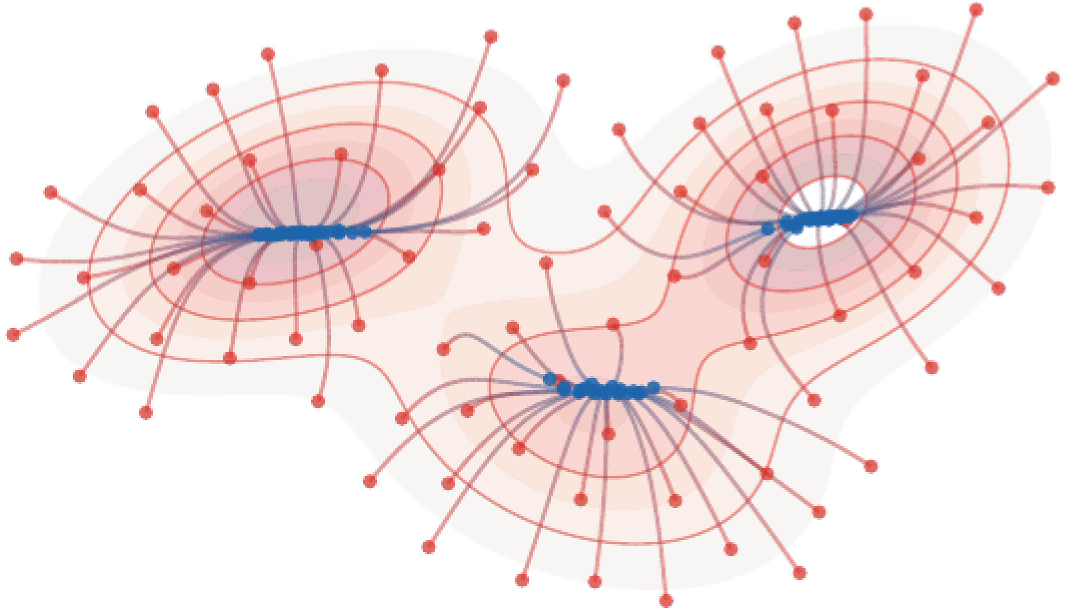
L2 attention and mean shift

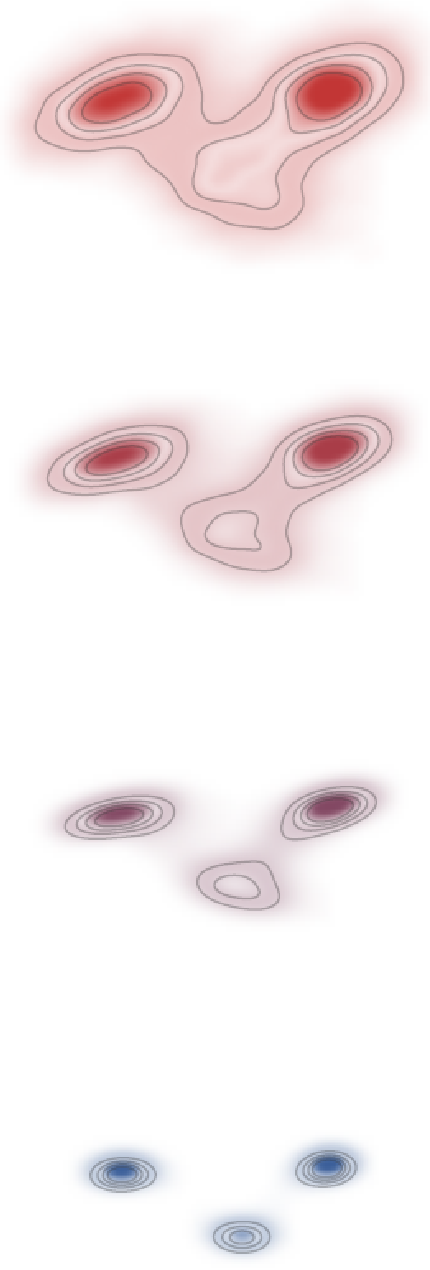
If attention scores use $-\|x-y\|^2/\varepsilon$, the continuum vector field resembles mean shift:

$$v_\alpha(x) = \frac{\int (y-x) e^{-\|x-y\|^2/\varepsilon}\, d\alpha(y)}{\int e^{-\|x-y\|^2/\varepsilon}\, d\alpha(y)}.$$

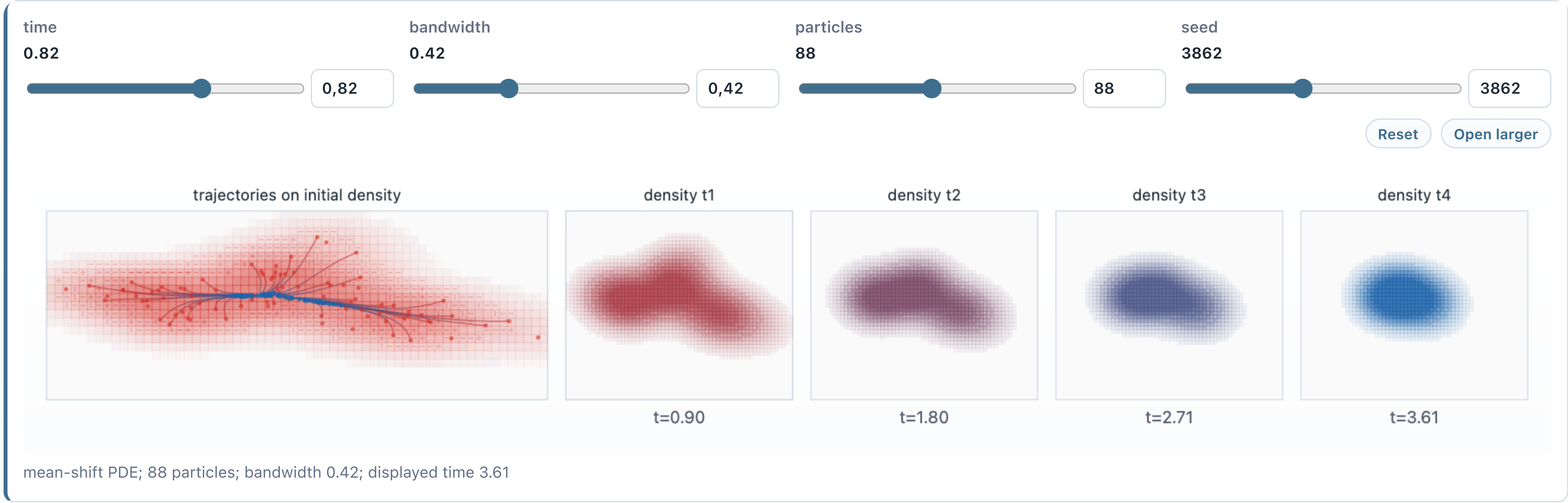
This moves particles toward local density modes.

Mean-shift PDE visualization





Live panel: mean-shift dynamics



Replacing dot-product attention by an L2 kernel reveals the mean-shift mechanism.

Gaussian case and clustering

For Gaussian mixtures, attention/mean-shift flows concentrate mass near modes. This gives a PDE-level explanation of clustering in token dynamics.

Message.
Transformer depth can be viewed as an evolution equation on tokens. In suitable limits, it becomes a nonlocal PDE for a probability measure.

Universality viewpoint

Deep transformers alternate equivariant interaction blocks. In the continuum viewpoint, they generate rich families of flows on empirical measures.

The qualitative message is:

- attention is an interaction kernel,
- depth is time,
- tokens are particles,
- expressive power comes from composing many equivariant flow steps.

Universality proof sketch

The reference deck used this as a final message: attention layers are not only a modeling heuristic, they can emulate rich equivariant computations.

Sketch.

Residual attention blocks build permutation-equivariant maps $(x_i)_i \mapsto (x_i + hA_i(X))_i$. By composing such blocks, one can localize interactions, separate token configurations, and approximate elementary equivariant operations. These operations generate dense families of continuous equivariant maps on compact sets; in discrete settings, related constructions underlie transformer universality results.

The PDE viewpoint interprets the same statement infinitesimally: expressive depth comes from composing nonlocal interaction vector fields.

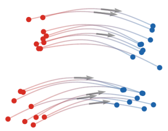
Interactive MyST figures

The project also contains browser-native interactive experiments mirroring several of these ideas:

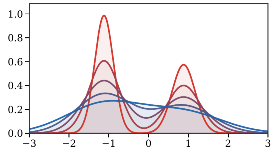
Gradient flows	Generative flows	Mean shift
JKO demo	flow demo	attention demo

Conclusion

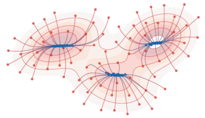
Takeaway.
Dynamic OT links maps, PDEs, particles, optimization, training, generation, and attention. The same continuity equation reappears under many names.



Sampling
flow matching and diffusion



Optimization
Wasserstein gradient flows



Learning
neurons, cells, and tokens